

MIT Lab Assignment – 03/09/26
Submission Due – 04/09/26, 11:59 PM.

A. For the following problems write the solving strategy with a small mathematical example, upload only one file (pdf) copy along with code.

1. Google PageRank – ranking billions of web pages using eigenvectors.
2. GPS Location Tracking – solving systems of equations from satellite distances.
3. Medical Imaging (CT/MRI Reconstruction) – rebuilding images from raw scan data.
4. Structural Engineering Vibrations – finding natural frequencies of bridges/buildings.
5. Wireless Communication (MIMO Systems) – ensuring independent signal transmission.
6. Economics & Market Equilibrium – solving supply-demand equations.
7. Computer Graphics (3D Transformations) – rotation, scaling, and rendering of 3D objects.
8. Cryptography & Coding Theory – encoding/decoding messages using matrices.
9. Epidemiology (Disease Spread Modeling) – predicting infection growth with transition matrices.
10. Air Traffic Control (Navigation Systems) – solving simultaneous equations for aircraft positioning.

B. Solve the following problems 1. By matrix factorization and 2. By basis concept of matrix.

11. Find the SVD of the following matrices.

$$M1 = [(1, -1), (0, 1), (1, 0)]$$

$$M2 = [(1, 1, 1), (-1, 0, -2), (1, 2, 0)]$$

12. Recommender Systems (Netflix, Amazon, Spotify).

Matrix factorization (like SVD, NMF) is used to predict user preferences (movies, songs, products) by reducing a large user–item rating matrix into latent feature matrices.

13. Machine Learning (Feature Spaces):

Data is represented in vector spaces; basis changes (like PCA basis) help reduce dimensionality and noise while keeping important variance.